Homework Answers

1. The Full Reset

Question: A file currently has `-rwxr-xr-x` permissions. You want to change it so the Owner has Read/Write, the Group has Read-only, and Others have No Access. What is the 3-digit numeric command?

Answer: The current permissions `-rwxr-xr-x` translate to `755` (Owner: `rwx`=7, Group: `r-x`=5, Others: `r-x`=5). To achieve the desired permissions:

- Owner: Read/Write (`rw-`) = `6`
- Group: Read-only (`r--`) = `4`
- Others: No Access (`---`) = `0`

Therefore, the 3-digit numeric command is `640`.

2. The Executioner

Question: You have a script with `644` permissions. You want to make it executable for everyone while keeping the Read/Write settings the same. What is the new 3-digit numeric code?

Answer: The current permissions `644` translate to (Owner: `rw-`=6, Group: `r--`=4, Others: `r--`=4). To make it executable for everyone while retaining the Read/Write settings, we need to add execute permission (`x`) to all categories:

- Owner: `rwx` = `7`
- Group: `r-x` = `5`
- Others: `r-x` = `5`

Thus, the new 3-digit numeric code is `755`.

3. The Secure Log

Question: A log file needs to be writable by the Owner and Group, but readable by everyone. What is the octal command?

Answer: To set the permissions as required:

- Owner: Writable (`rw-`) = `6`
- Group: Writable (`rw-`) = `6`
- Others: Readable (`r--`) = `4`

The octal command is `664`.

4. Binary to Octal

Question: If a file's binary permission string is `110 100 000`, what is the corresponding `chmod` numeric command?

Answer: Converting the binary permission string to octal:

- Owner: `110` (binary) = `6` (octal)
- Group: `100` (binary) = `4` (octal)
- Others: `000` (binary) = `0` (octal)

The corresponding `chmod` numeric command is `640`.

5. The Mystery Code

Question: What happens to a file if you run `chmod 000 secret.txt`? Can the Owner still `cat` the file without using `sudo`?

Answer: Running `chmod 000 secret.txt` removes all permissions (read, write, execute) for the owner, group, and others. This means no one, including the owner, can read, write, or execute the file. Consequently, the owner **cannot** `cat` the file without using `sudo`, as they would lack the necessary read permissions.

6. File Creation

Question: Create a file named “Audit.txt” and assign it the following permission: `rwxr-x---`

Answer: First, create the file `Audit.txt`. Then, convert the symbolic permission `rwxr-x---` to its numeric (octal) equivalent:

- Owner: `rwx` = `7`
- Group: `r-x` = `5`
- Others: `---` = `0`

The numeric permission is `750`. The commands would be:

```
touch Audit.txt
chmod 750 Audit.txt
```

7. Numeric to Symbolic

Question: What is the equivalent of a `741` permission in symbolic notation?

Answer: Breaking down the numeric permission `741`:

- Owner `7`: `rwx` (read, write, execute)
- Group `4`: `r--` (read only)
- Others `1`: `--x` (execute only)

The symbolic notation is `rwxr-x--x`.

8. Numeric to Symbolic

Question: What is the equivalent of a `500` permission in symbolic notation?

Answer: Breaking down the numeric permission `500`:

- Owner `5`: `r-x` (read, execute)
- Group `0`: `---` (no access)
- Others `0`: `---` (no access)

The symbolic notation is `r-x------`.

9. Symbolic to Numeric

Question: What is the numeric equivalent of `-r-xrw-r--`?

Answer: Breaking down the symbolic permission `-r-xrw-r--`:

- Owner `r-x`: `5` (read, execute)
- Group `rw-`: `6` (read, write)
- Others `r--`: `4` (read only)

The numeric equivalent is `564`.